ALUMNI CONNECT AND NETWORKING PORTAL

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Abstract:- In today's competitive job market, students often face challenges in securing industry referrals and guidance, while colleges struggle to maintain an active connection with their alumni. The Alumni Connect & Networking Portal is a comprehensive full-stack web application designed to bridge this gap by connecting current students, alumni, and placement cells through a secure and structured online platform. The main goal of this platform is to bridge the gap between alumni and students for mentorship, job referrals, and industry insights. Students can request referrals for specific job roles, attempt quizzes to validate their skills, and receive timely notifications about job openings shared by alumni. Alumni, on the other hand, can post job referrals, connect with students, and guide them towards relevant opportunities, ensuring that their support reaches the right candidates. An admin module ensures the platform's credibility by verifying and approving alumni profiles an monitoring all activities to maintain transparency and security. The platform implements a secure authentication system with role-based access control to differentiate between students, alumni, and admin users. The project uses React JS for an interactive and user-friendly frontend, Spring Boot for a robust and scalable backend, and MongoDB for flexible and efficient data storage. By integrating all stakeholders through a centralized digital channel, this solution addresses the common issue of students missing out on job referrals and placement teams struggling to track alumni contributions. Overall, this portal aims to strengthen the bond between colleges and their alumni network, creating a reliable ecosystem that supports students career growth and enhances placement outcomes.

Keywords:- Active connection, Mentorship, Skill evaluation, Role based access, Secure, Guidance, Referrals, Centralized digital Solution.

I.INTRODUCTION

An Alumni and their success was a big asset for any college. Alumni play a crucial role between colleges and job industry. Their guidance through mentorship, career development and referrals for the students adds most valuable success rate for student development and the respective college. Despite of this,many colleges have designed their specific platforms for significant guidance and mentorship from alumni s to students and they may lack in proper communications among alumni and students and also structured way of maintenance of the platform by the placement team which may results in missed opportunities and industry insights for students. At a certain level, many students face challenges in building active connections among alumni and on other side placement team(Admins) also often struggle in maintaining updated activities of alumni and real time achievements. This prevents institutions from leveraging the collective experience of their alumni network.

To overcome all these challenges, a digital centralized platform called Alumni Connect and Networking Portal has been designed to strengthen the relationships between alumni, student and placement departments. It provides verified and secure environment for all users. The platform integrates multiple features to enhance collaboration and employability.

Alumni Connect and Networking Portal aims to provide referral based opportunities and professional career guidance. This portal also improves interactions between alumni and student by involving personal real time conversation between one to one. Additionally, it offers transparent monitoring tools for administrators to assess alumni engagement and track the impact of their activities and contributions. It helps to maintain a sustainable and collaborative friendly environment between institutions and their alumni community by ensuring valuable guidance reaches to a skilled person from an active alumni.

Apart from support system and communications it also involves in promoting job roles and job openings along with required skills directly on a portal enabling students to access trusted opportunities. It can also helps alumni to refer limited number of students in a shorthand pressure. Students can maintain a skill profile on the platform, showcasing their technical abilities, projects, certifications, and achievements, which helps alumni and recruiters identify potential candidates more efficiently. This portal also provides some quiz modules for students on which alumni can filter deserving students which can be monitored by placement teams.

In short, Alumni connect and networking portal not only bridges the gap between alumni and students, it also provides dynamic environment by enhancing continuous interactions between alumni and students, continuous networking and career advancement. It integrates all monitoring and guidance activities, job referrals, skill evaluation and interactions within one platform.

II. LITERATURE REVIEW

Based upon some literature survey We know a little more from some journals to work on this design. We have collected many journals that are applicable to our design that will be more useful to us in preparation part of our design. I will give a gist of the journals that we want to use in the development of this web operation.

To ensure the successful integration of all proposed features within our project and to understand some foundation of such initiatives a review of existing research and systems related to alumni engagement, mentorship, and smart networking tools is essential. The following literature survey presents studies that highlight the significance of alumni networks and the technological approaches used to enhance alumni- student interaction.

a. Alumni Connect Forums

Barudwale et al. had explored the concept of alumni connect forums in the year 2023 which relies on the approach of mentorship and networking tools. It aims at the strengthening of the relationships between alumni and educational institutions. This emphasizes the use of various career guidance programs and interactive events and also some digital networking tools that are accessible for knowledge sharing and career guidance support among students and alumni.

This research concludes that though the alumni interactive programs and supporting network tools helps students to develop but the long term success of institution and consistent development mainly depends on the having consistent engagement between alumni and students. This advise us the importance of structured collaboration or communication platforms that encourages continuous interaction and relations between participants.Chi, Jones & Grandham researched on mentoring via smart alumni systems in the year of 2012 to enhance institutional support mechanisms.

b. Mentoring via Smart Alumni Systems

They worked on the methods smart systems and mentorship. It involves technology driven mentoring among students and alumni by paring students with one skilled and experienced alumni for collaboration and supporting. It was one of the successful approach by organizing the way of mentoring students. It improves the quality of interactions and monitoring. However, it was succeeded but the efficiency of such systems largely depends on active participation of both alumni and students.

This research concludes that Structured support enhances mentorship but success relies on active participation.

c. Student & Alumni Network site

Golloh focused on Social Networking Site for Students and Alumni in year 2018. It mainly focused on how digital platforms enhances the collaboration and communication among educational institutions. It deeply focused on the alumni-student networking sites. It highlighted the point that they are the most impactful tools for building a strong connection between them. It also provides professional dashboard and connections.

Moreover, It also mentioned that these platforms are under the issues of proper monitoring because of lack of structure ability which may leads to reduced engagements. The research on this concludes the importance of well maintaining and actively managed system to establish meaningful interactions over participants.

d. Student-College-Alumni Relationships

McAlexander & Koenig had worked on the area Student-College-Alumni Relationships in the year of 2001. They mainly focused on the aspects of this through Surveys and relationship analysis. The research was concluded on points that active participation of students demonstrates the vital role even after completing their graduation. This study summarised that fostering positive relation during student phase will maintain healthy and active connections and also contributes long term success to the institution.

The reviewed studies gives information an effective alumni engagement system should combine structured communication, technological integration, and strong relationship-building. Existing research emphasizes that while online platforms and mentorship initiatives help connect students with professionals, many lack proper organization and consistent participation. It is also evident that successful alumni networks depend not only on digital tools but also on continuous interaction, trust, and institutional support. These insights form the basis for developing the proposed Alumni Connect & Networking Portal, which aims to create a structured, interactive, and sustainable environment for collaboration between students, alumni, and placement teams.

III. RELATED WORK

Before the introduction of the Alumni Connect & Networking Portal, student referrals typically took place through informal and unstructured channels. Most students relied on personal contacts, social media platforms like LinkedIn, or word-of-mouth communication to reach out to alumni working in various organizations. In many cases, placement cells maintained limited alumni contact lists or manually connected students with past graduates for potential job opportunities. However, this process was treated as inefficient, as not every student has the right contacts or knew whom to approach. There was also no proper system to verify referrals, monitor engagement, or track outcomes. As a result, many deserving students missed valuable opportunities due to the unavailable of a centralized and transparent platform for alumni-student interaction and referral management.

The absence of a centralized system also created difficulties in maintaining transparency and verifying the authenticity of referrals. Since most interactions happened outside institutional oversight, there was no way to ensure whether the referrals provided were genuine or tracked effectively. Placement cells had minimal control over the referral process and were not able to monitor how many students grew from alumni connections. This unstructured approach not only reduced the performance of alumni engagement but also prevented the institution from leveraging its alumni network to its full Potential.

With the implementation of alumni connect, all the referrals were trust worthy and approved guidance approach and the entire referral process has become structured, efficient, and transparent. With beginning of this, Every student and alumni gets to register into the portal with an verified usernames or emails and passwords and the alumni must be authenticated to get access over the portal to post referrals and maintain connections. Here the administrations must work to verify the details of alumni and approve them.

The Student needs to create their profile which will be visible to both alumni and placement team with uploading their resume and other certifications they had done. Once completed the process, students can view a dynamic dashboard with their details along with job postings and referral oppurtunities and the related skills for that role and etc.. The student can connect alumni for further communications and guidance with real time chats. They can search for right alumni by filtering on basis of company they are working. Students can also evaluate their knowledge on a particular subject by attempting quiz. Students will also get notify about the approval or rejection of their referral from admins. Our portal also promotes the upcoming technical events for students and they can easily register for specific interested event from the portal directly.

On the other side of portal, The Alumnus will also needs to register for this portal and they must get confirmation from the administrations which can refrain from the misconduct or improper actions. Here, the main key of security is the manual approval of alumni profile by placement team. It can be done with the proper verification of personal profile and college data history.Once alumni get approval, they can logs into the portal and they can view students profiles and their skills. If they had any job referral opportunities they can be posted in the portal and the alumni are responsible for the selection of right candidates based on the requirement.

They can filter students based on resumes, quiz performance or skills they mentioned. Alumnus have access to view the reports of quizzes attempted by the students if any. They can search and connect

students and can also communicate with them for career guidance, mentorship and skill development. The communication provides one to one way of communication which provides secrecy of data and conversations.

IV. PROPOSED SYSTEM

The proposed system provides the new way of providing referrals for students and guiding them and monitoring their activities. It is the single platform which brings all the participants such as students, alumni and placement team. Referenced on some existing systems our platforms inherits more advanced features for both student and alumni.Our Alumni Connect & Networking Portal provides user centered and dynamic web portal. It provides role based web pages for participants. The following shows the proof of our work

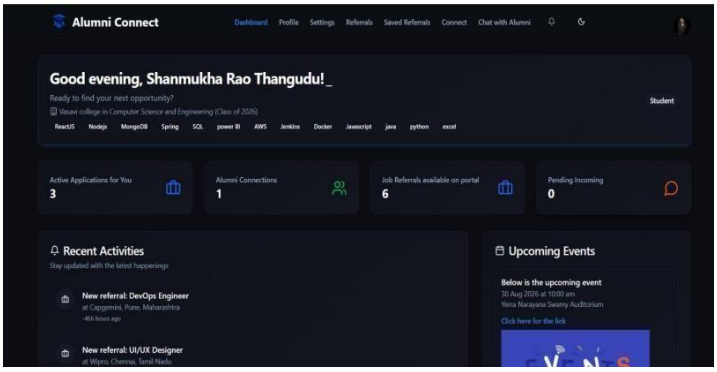

fig1: student based dashboard

The above figure shows the student based dashboard. It shows recent activities updated in portal and upcoming events and different activities to perform by the students. After successful login a student can view dynamic dashboard which maintains updated functionalities. Student can connect with alumni filtered by companies and also it shows total number of alumnus availble to connect.

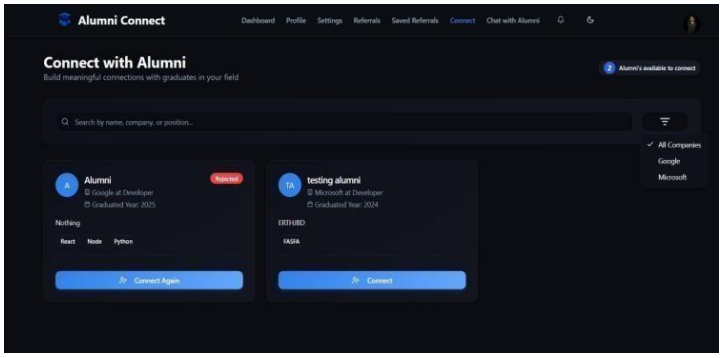

fig 2: Alumni finding page for students

Student can view different job referrals for specific position posted by Alumni. He/She can view full details of referral and can apply if they are interest and also can attempt assessment for that position through can evaluate their performance.

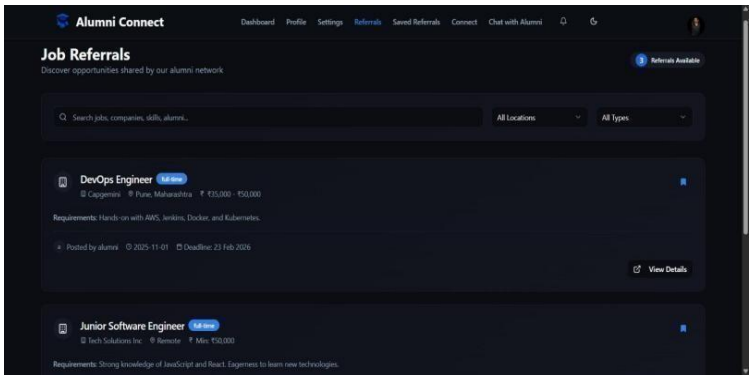

fig 3: Student referral access page

On the other side of portal i.e Admin. Admin can view monthly platform Activity on an average and user statistics based on graph. Admin can see referral success rate based on how many students got hired from the portal. Admin can view total number of users, referrals and can monitor all the activities. Overall in our proposed system, The system provides a secure and interactive web application where users can register, create profiles, and join communities based on graduation year, domain, or interests. The platform supports real-time updates about events, job opportunities, and alumni achievements. Admins can monitor user activities, approve registrations, and manage announcements to ensure authentic and relevant communication.

The Alumni Connect system also incorporates a referral and mentorship feature, enabling alumni to refer students for job roles or internships. A search and filter functionality allows users to easily locate alumni from specific batches, departments, or industries. The system ensures data privacy, security, and scalability.

This proposed system aims to create a long- term sustainable network that benefits both the institution and its members by encouraging continuous interaction, collaboration, and career support.

VI. DATASET DESCRIPTION AND EXPERIMENTAL SETUP

The dataset utilized in this study was obtained from a prototype version of an Alumni Connect and Networking Portal, which was developed as part of an academic project for system design and performance evaluation. The data comprises a combination of synthetically generated records and real-time simulated interactions to closely resemble real-world usage scenarios.

The dataset includes profiles of 300 students, 120 alumni members, and 85 job referral records, spanning application areas such as software development, data science, and core engineering disciplines. Student-related information consists of academic background, technical skills, certifications, quiz performance, and resume-related metadata. Alumni records include details such as year of graduation, current organization, job designation, and past referral activities.

In addition to profile data, the dataset captures interaction-based records such as quiz attempts and mentorship engagements, which were used to evaluate system functionality and user activity tracking. All personal identifiers were anonymized to ensure privacy and ethical use of data.

The data was organized using MongoDB collections, allowing flexible schema design and supporting scalability for future expansion in terms of users and interaction logs. The dataset was used exclusively for system validation, feature testing, and performance analysis of the proposed portal.

TABLE 1: DATASET DESCRIPTION

DATASET COMPONENT	COUNT
Student	1300
Alumni	4000
Job Referrals	100
Quiz Attempts	800

VI. SYSTEM ARCHITECTURE

The System Architecture of our proposed system is a three-tier architecture which consists of Frontend, Backend and Database which ensures efficiency, scalability and security. Different layers can be explained as follows :

a. User Layer :

It supports three different users Students, Alumni and Admin.

Students: Can register or login, update their profile, seek mentorships, attempt quizzes, apply referrals, upload their resumes and other certificates and can also participate in events.

Alumni: Can register or login, View students profiles, post job openings and guide students.

Admin: Can view updates, monitor activities, verifies posts, handle users.

b. Frontend Layer :

The Frontend is built using React.js, ensuring a dynamic and responsive user interface. It contains three major components:

UI Layer: Responsible for displaying pages, dashboards, and interactive forms.

Components & Pages: Modular React components that render specific features (e.g., login, job posting, mentorship portal).

API Client: Connects the frontend to the backend using RESTful APIs for sending and receiving data securely. This layer provides an engaging and user-friendly experience while maintaining separation from the backend logic.

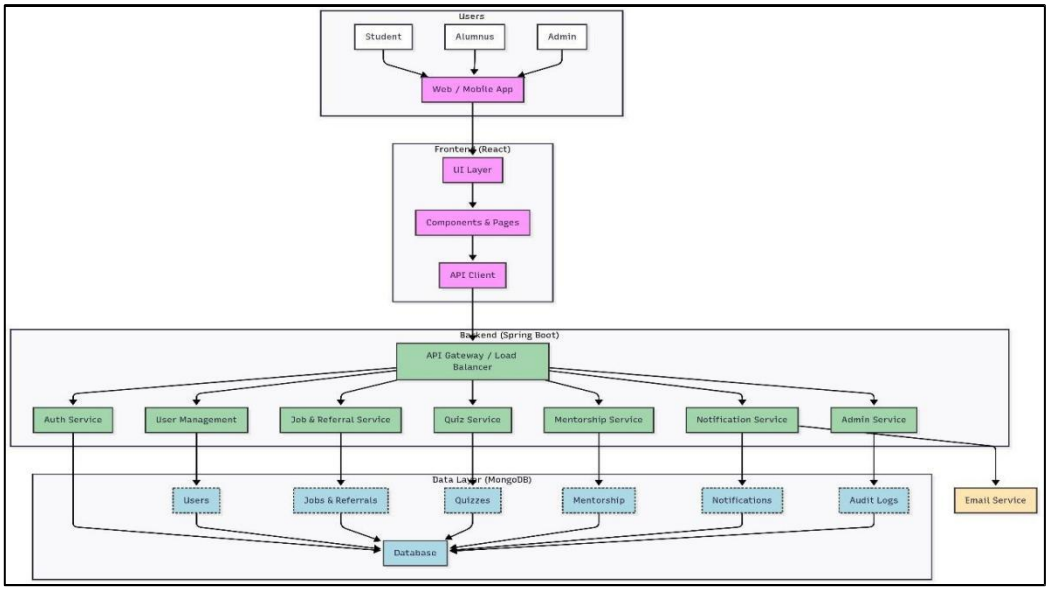

fig 4: System Architecture for Alumni Connect & Networking Portal

c.Backend Layer :

The backend layer provides different types of micro services. It ensures flexibility and modularity. It provides following services :

Auth Service: Manages user authentication and authorization using secure JWT-based login.

User Management Service: Handles profile creation, updates, and user-related operations.

Job & Referral Service: Allows alumni to post job opportunities and provide student referrals.

Quiz Service: Manages quizzes or learning modules to enhance student–alumni engagement.

.Mentorship Service: Helps to connect students and alumni for career support and guidance.

Notification Service: Sends system updates, event alerts, and connection notifications to users.

Admin Service: Provides administrative control for user verification, report generation, and monitoring activities.

d. Data Layer :

The Data Layer stores and manages all application data securely using MongoDB.It contains collections such as:

Users – Stores alumni, student, and admin profiles.

Jobs & Referrals – Contains job postings and referral data.

Quizzes – Stores quiz questions and results.

Mentorship – Tracks mentor– mentee connections and sessions.

Notifications – Logs system and user notifications. Audit Logs – Records all major user and system activities for tracking and security.

This NoSQL database provides scalability, flexibility, and high-speed data access.

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VI. RESULTS AND DISCUSSION

Unlike conventional alumni networking platforms that mainly emphasize informal social interaction, the proposed system extends functionality by incorporating referral validation, quiz-based skill assessment, and administrative supervision. Most existing systems do not provide a structured mechanism to track referrals or analyze their outcomes, which limits accountability and effectiveness. The present work addresses these limitations by offering a centralized framework that supports authenticated referrals, performance-based candidate evaluation, and continuous monitoring, thereby improving reliability and transparency. The implementation of the Alumni Connect and Networking Portal demonstrates notable improvements in managing alumni–student interactions and referral processes. In comparison with traditional manual methods or unstructured social media platforms, the proposed system facilitates organized communication, verified referral handling, and centralized administrative oversight.

The integration of quiz-based skill assessment assists alumni in identifying eligible candidates more efficiently, thereby reducing the effort required for preliminary screening. Furthermore, the presence of administrative verification mechanisms enhances data authenticity and prevents misuse of referral privileges. Experimental evaluation and functional testing indicate reduced processing time for referral validation and increased participation levels among registered users.

Overall, the system shows clear advantages in terms of transparency, accessibility, and user engagement when compared with existing alumni interaction approaches, highlighting its suitability for academic institutions seeking structured and reliable alumni networking solutions.

VII. CONCLUSION

In the light of the findings of the study, the following conclusions were drawn:

The Alumni connect and networking portal ensures several functionalities availble to both students and Alumni. It includes

a. Role based logins for alumni and studnets.
b. Students can view referral opportunities and apply them if eligible.
c. Alumni can approve right students for referrals.
d. Students can attempt quizzes for skill evaluation.
e. Students can share their resumes and completed certifications.
f. Students and Alumni can have a private conversations individually.
g. All participants can view different upcoming events.
h. Placement team can approve or reject alumni logins and can also view students performance.
i. Placement team can monitor all the activities on the portal.

Thus all the above advanced features were provided in our alumni connect an networking portal. This platform not only upgrades collaboration and career guidance but also strengthens the bond between alumni and their alma mater, promoting a culture of continuous learning and mutual support. In the future, the system can be further enhanced with AI-based recommendations and advanced analytics to make the alumni–student interaction more personalized and impactful.

VIII. DECLARATION

I hereby declare that the paper report named “Alumni Connect & Networking Portal” is my proof of original work that has been completed under the assistance of SD.ARIEF sir. The content of this report is based on my research and complete understanding of the project and subject, using various literature sources, tools and techniques.

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